

DISEASE DETECTION SYSTEM FOR SRI LANKA'S COMMON GROWN CROPS

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Crop diseases pose a significant problem for the agricultural sector in Sri Lanka, frequently leading to substantial harvest losses. Traditionally, diagnosing plant diseases requires expert intervention, a process that is costly, time-consuming, and often inaccessible to rural farmers. To address this gap, this project proposes an automated disease detection system utilizing a custom-built Convolutional Neural Network (CNN) to instantly and accurately identify crop diseases from leaf photographs.

The model was trained on a refined version of the New Plant Diseases Dataset, focusing specifically on 25 disease categories across six crops commonly cultivated in Sri Lanka: Corn, Grape, Bell Pepper, Potato, Strawberry, and Tomato. The custom CNN architecture, developed from scratch using TensorFlow and Keras, consists of five sequential convolutional and max-pooling blocks designed for progressive feature extraction, integrated with dropout regularization to mitigate overfitting. The model was compiled using the Adam optimizer with a reduced learning rate to ensure stable convergence.

Following 40 epochs of training, the model demonstrated exceptional performance, achieving a training accuracy of 99.19% and a validation accuracy of 96.12%. The minimal gap between these metrics confirms the effectiveness of the regularization techniques applied. To facilitate practical user interaction, the trained model was successfully deployed through a Streamlit web application, enabling real-time disease prediction and confidence scoring from uploaded leaf images. These findings highlight the viability of applying accessible deep learning tools to support localized agricultural disease management.

Keywords: Crop Disease Detection; Convolutional Neural Networks; Deep Learning; Sri Lanka Agriculture; Streamlit.